Getting Started with ActiveMQ

Scott Cranton
Principal Solution Engineer
scranton@fusesource.com

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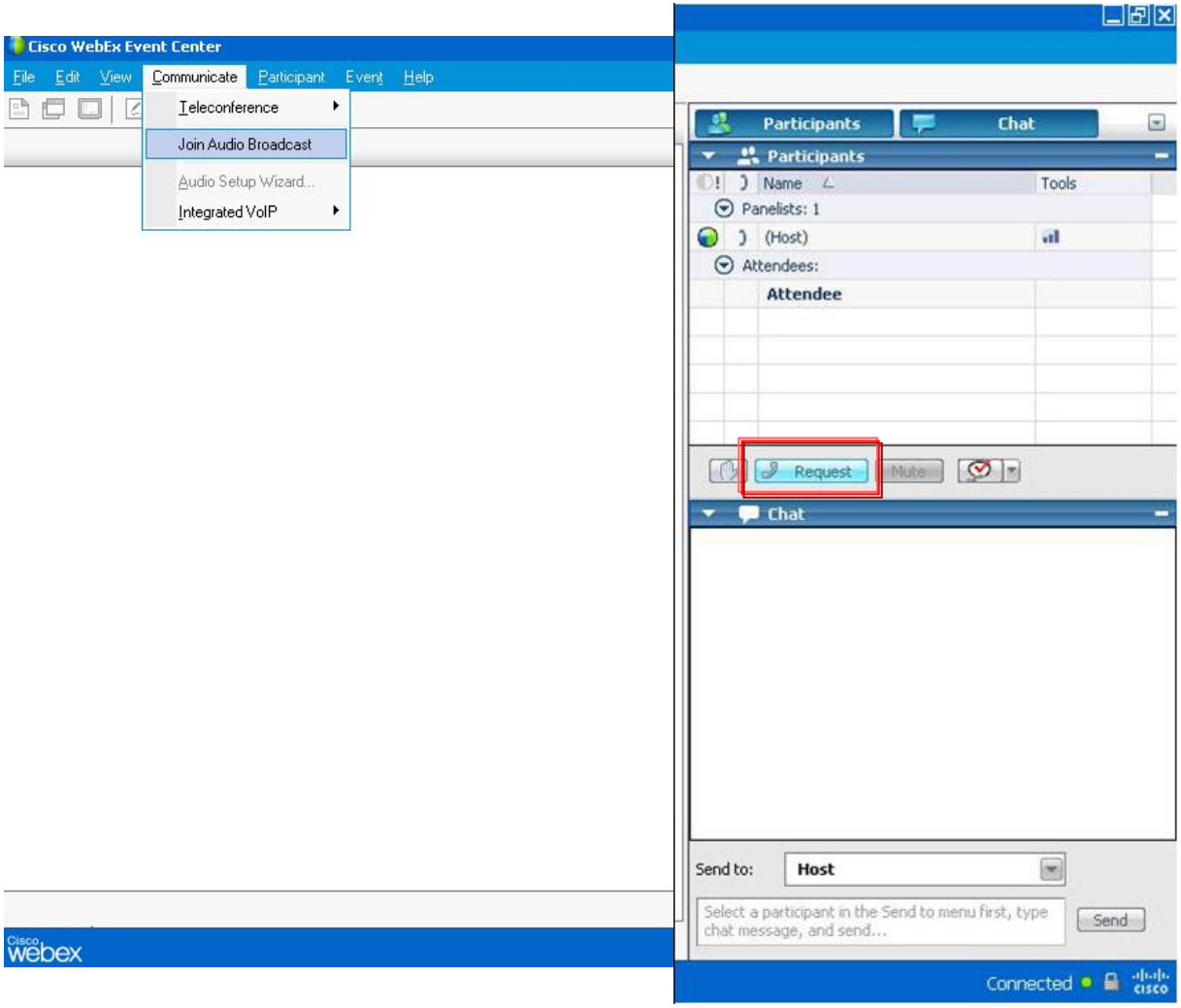

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Agenda

- Who's FuseSource?
- ActiveMQ Overview
  - Core capabilities
  - Managing client connections
  - Managing persistence
  - High availability
  - Network of brokers
- Demo
  - Walk through install
  - Start/Stop with alternative configuration
  - Management through JMX
  - High availability: failover and back

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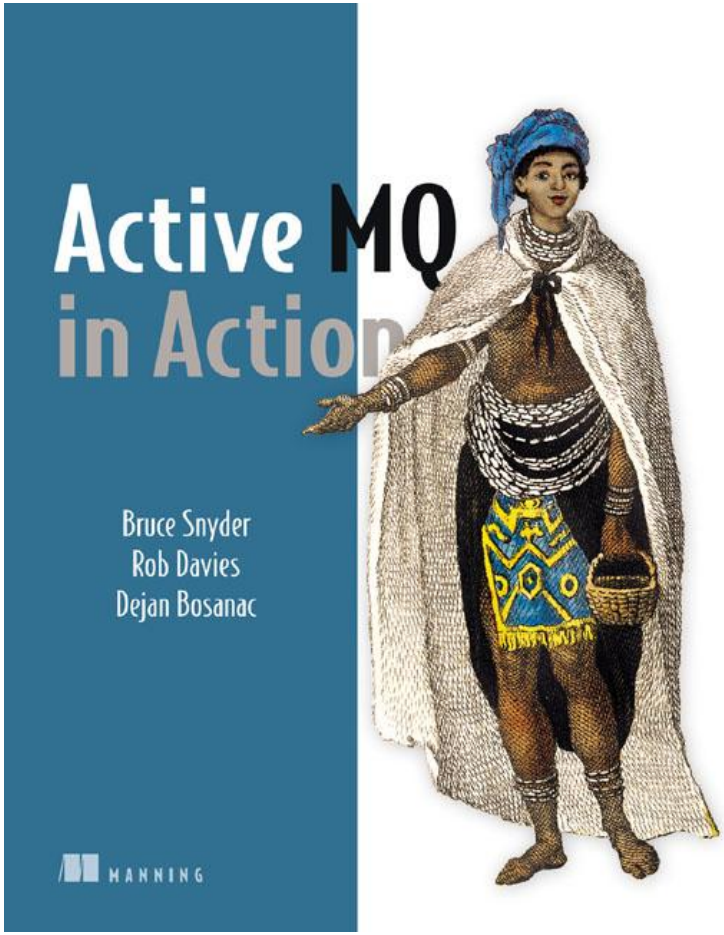
Book Giveaway: ActiveMQ in Action



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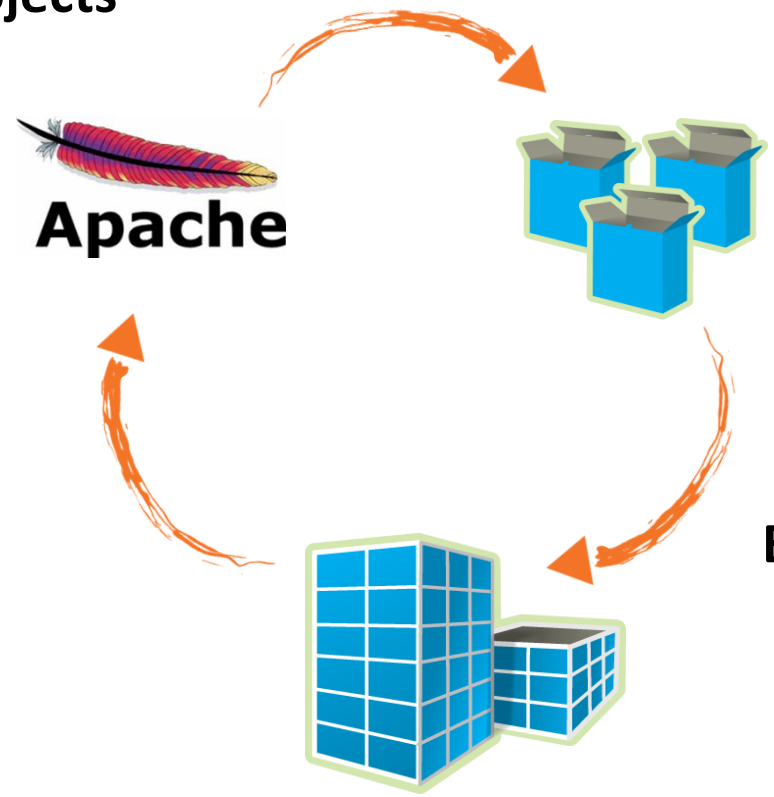
The Leaders in Open Source Integration and Messaging

Team behind the projects

- Leaders at Apache
- Product roadmaps
- Code contributions



Productized distributions

- Integrated
- Tested
- Tooling

Enterprise support

- Subscriptions
- Training
- Consulting

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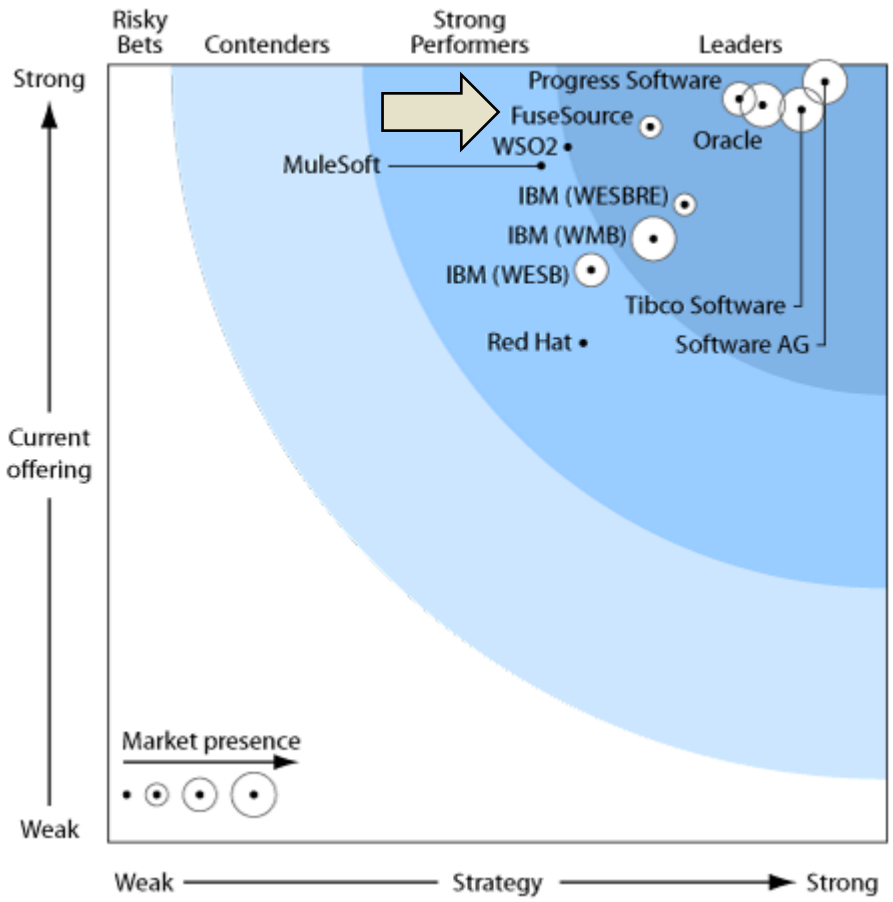
Forrester Wave Report Q2 2011: Fuse ESB is a "Leader"

- FuseSource placed in "Leader" category in company with large, established vendors
- One of few open source solutions considered for this report
- Highest ranked open source solution



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FuseSource : Open Source Integration for the Enterprise

Cost-effective, Proven, Enterprise-class Solutions

- Same Apache code, but tested, productized and supported
- Business-friendly, open source (Apache) license
- Over 25 active Apache committers on staff

Apache Project	FuseSource Product
Apache ServiceMix	Fuse ESB ESB with OSGi and JBI
Apache ActiveMQ	Fuse Message Broker Reliable messaging: Java JMS, C++ and .NET
Apache CXF	Fuse Services Framework SOAP, XML, and REST web services
Apache Camel	Fuse Mediation Router Enterprise integration Patterns

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FuseSource : Team that Wrote the Code

No one knows the code, or influences the projects at Apache more than FuseSource:

- Co-founders and PMC members of ServiceMix, ActiveMQ, Camel, …
- Over 25 active committers on 11 Apache projects

Guillaume Nodet

James Strachan

Rob Davis

Hiram Chirino

Jon Anstey

Gary Tully

Dejan Bosanac

Gert Vanthienen

Willem Jiang

Claus Ibsen

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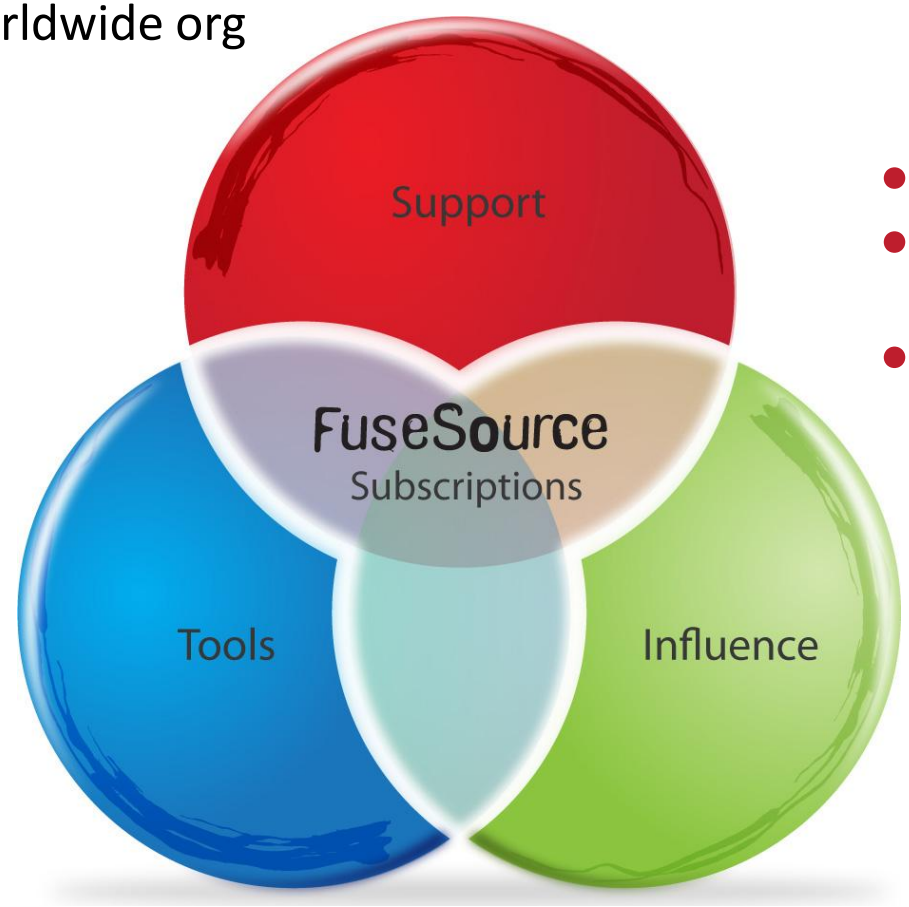
FuseSource Subscription = Long Term Success

Support

- From the project leaders
- Enterprise-class
- Worldwide org

Tools

- Pilot projects
- Development
- Deployment



Influence

- Product knowledge
- Effect product direction
- Partner with the developers

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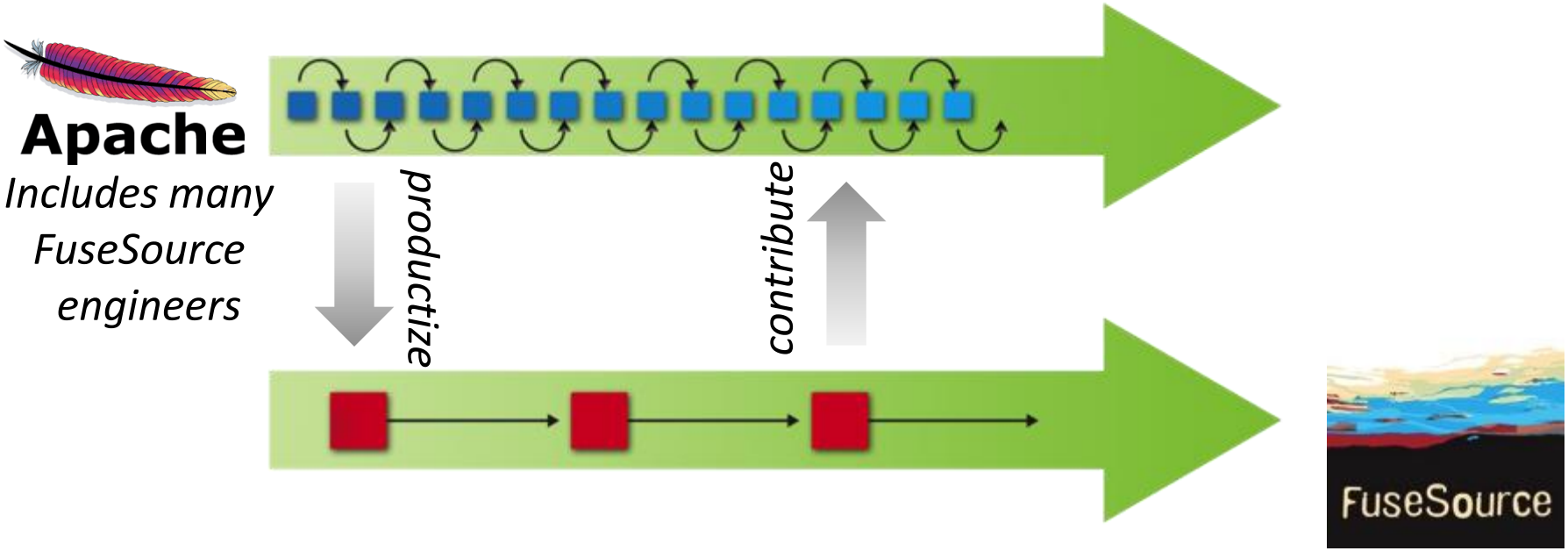
FuseSource Subscription : Alignment with Apache

FuseSource includes the leaders & founders who drive the projects

- No one knows the internals of the projects better
- FuseSource has access to product road maps
- Customer patches are contributed to Apache
- Customer feedback drives project direction



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What is Apache ActiveMQ?

- Top level Apache Software Foundation project
- Wildly popular, high performance, reliable message broker
  - Supports JMS 1.1; adding support for AMQP 1.0 and JMS 2.0
  - Clustering and Fault Tolerance
  - Supports publish/subscribe, point to point, message groups, out of band messaging and streaming, distributed transactions, …
- Myriad of connectivity options
  - Native Java, C/C++, and .NET
  - STOMP protocol enables Ruby, JS, Perl, Python, PHP, ActionScript, ...
- Embedded and standalone deployment options
  - Pre-integrated with open source integration and application frameworks
  - Deep integration with Spring Framework and Java EE

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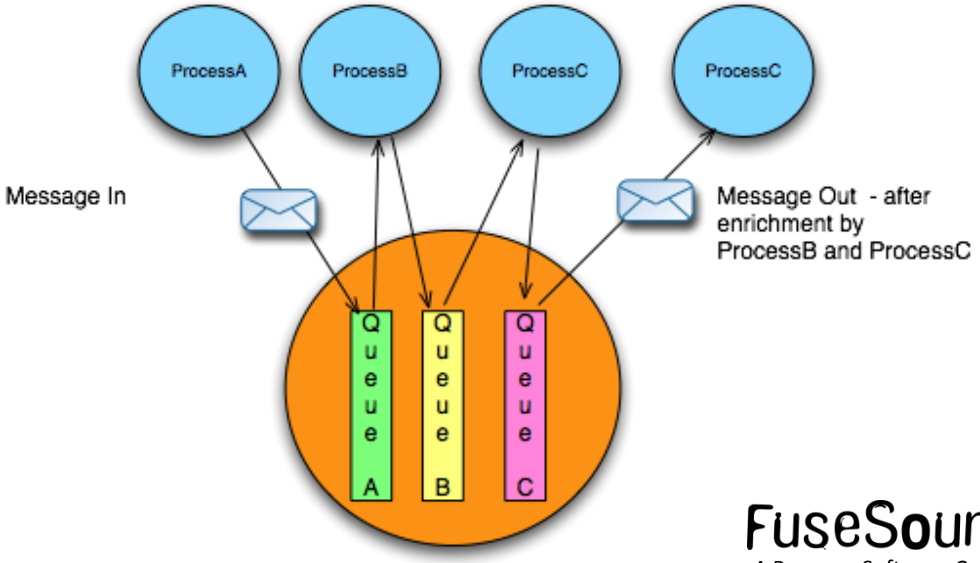
Why use Messaging?

- Reliable remote communication between applications
- Asynchronous communication
  - De-couple producer and consumer (loose coupling)
- Platform and language integration
- Fault tolerant - processing can survive Processor outage
- Scalable - multiple consumers of each queue
  - Distributes processing



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Message Channels and Routing

- Message Channels
  - Named communication between interested parties
  - JMS calls them ‘Destinations’
- Can fine-tune message consumption with selectors
- Can route a message based on content

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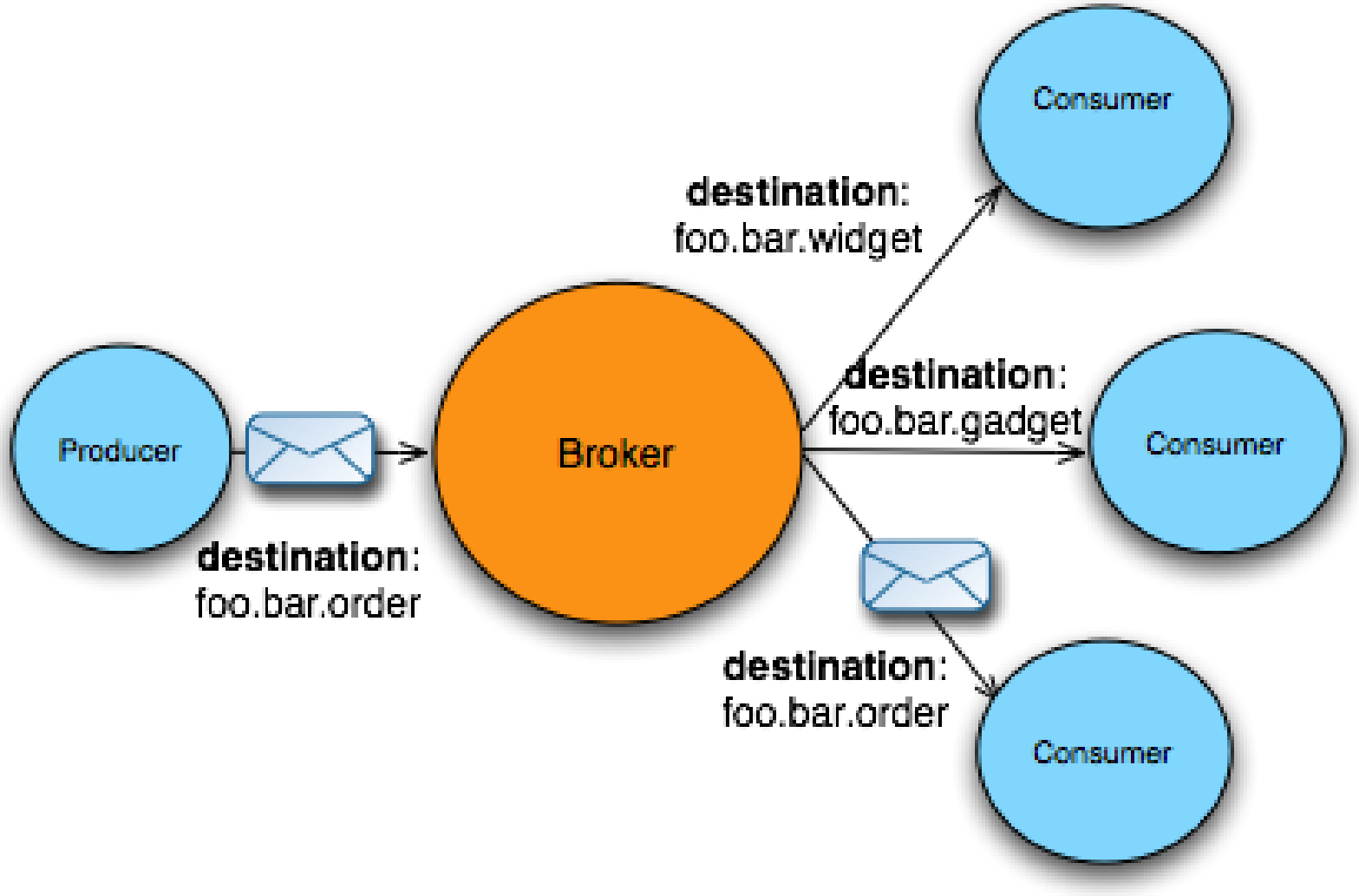
Message Channels = JMS Destinations



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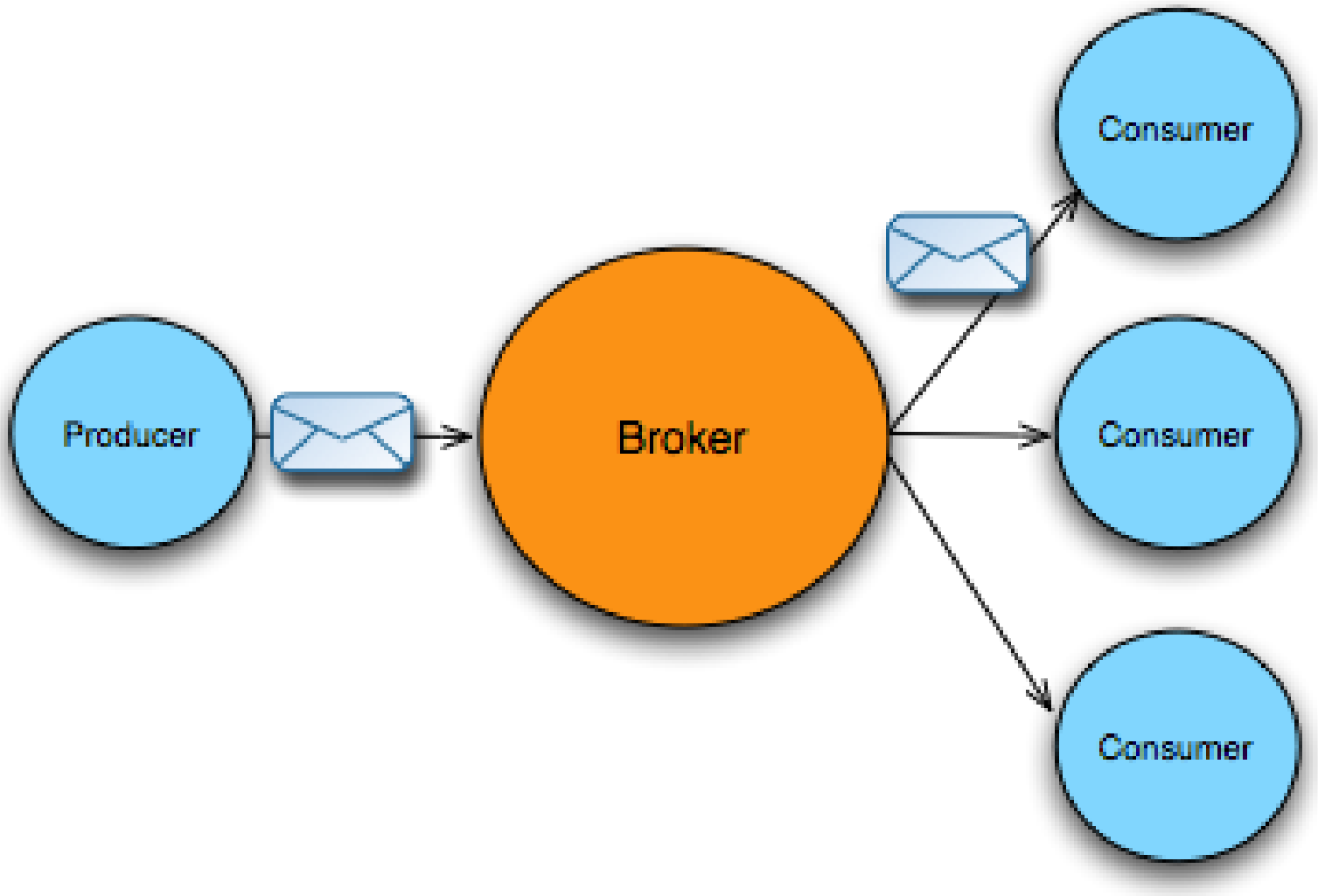
Point-to-Point Channel : JMS Queues



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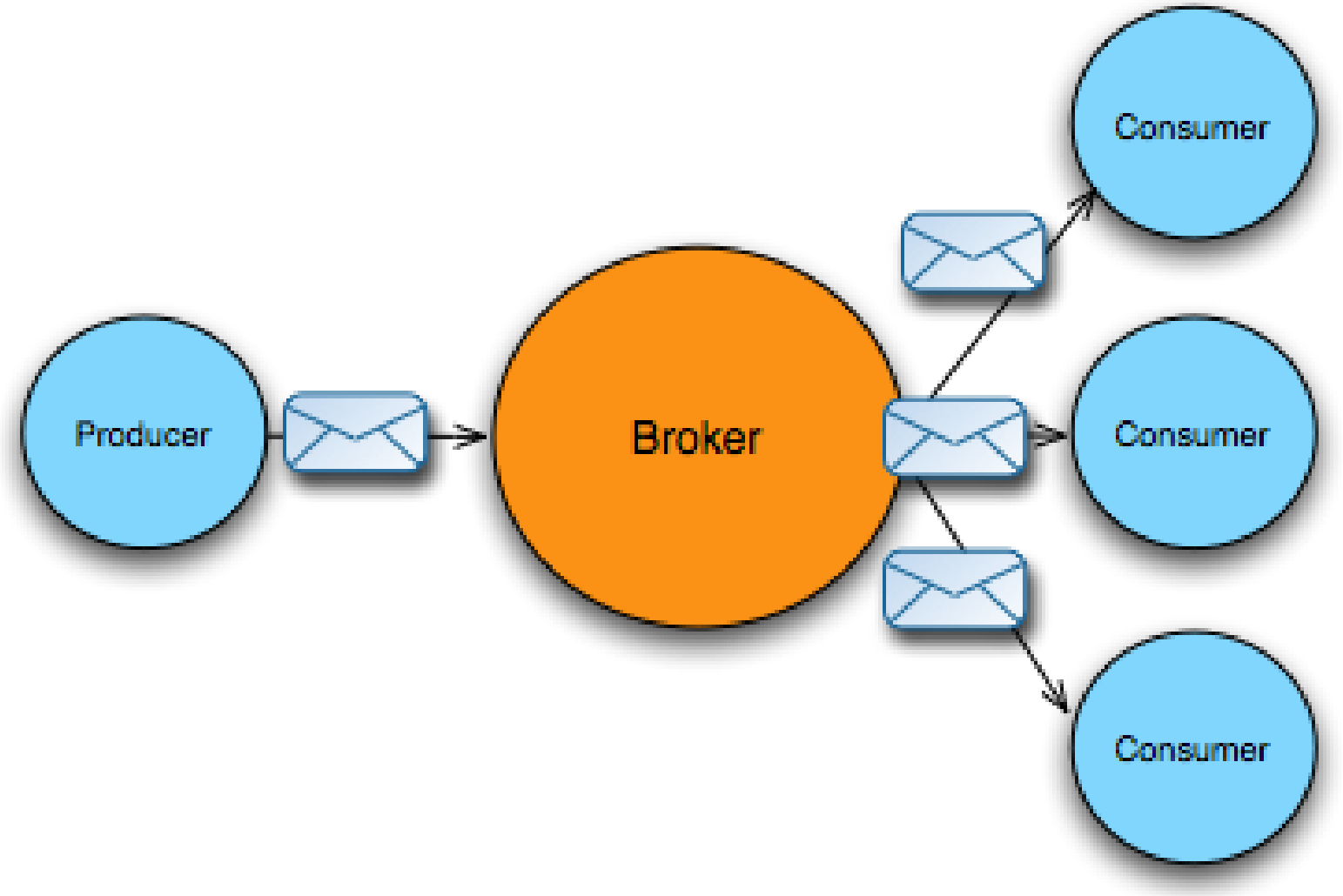
Publish/Subscribe Channel : JMS Topics



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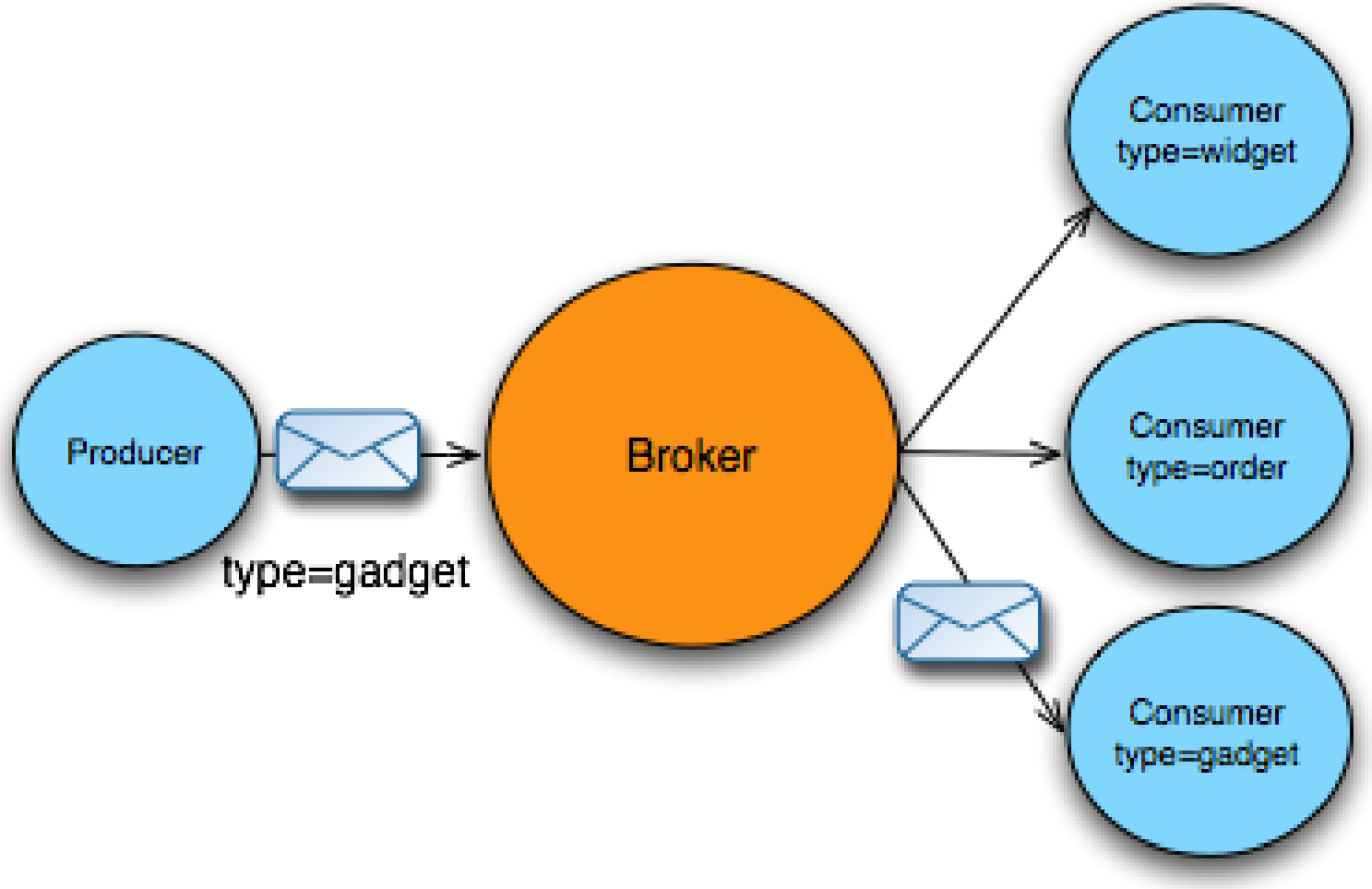
Message Routing : Selectors



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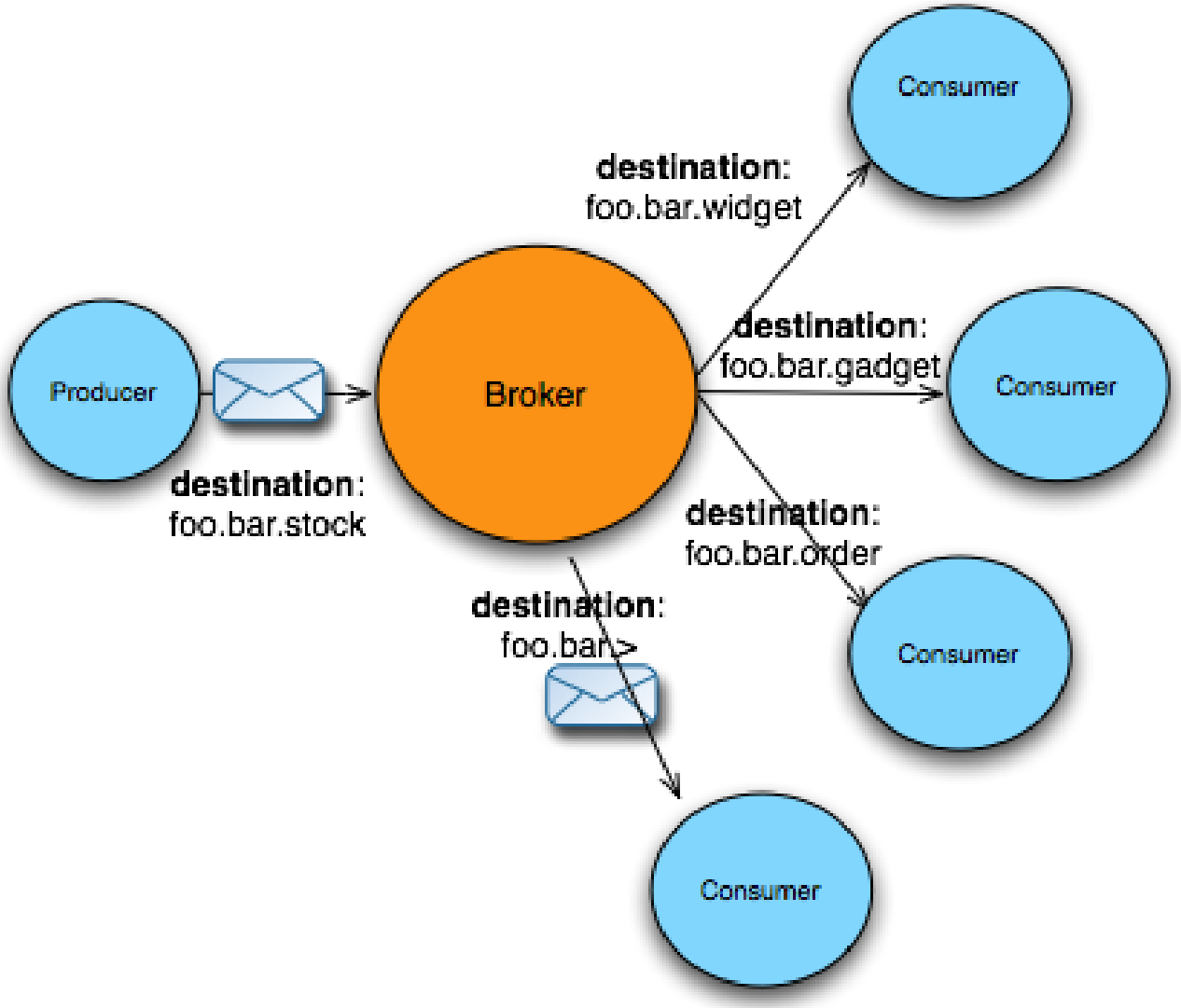
Message Routing : Destination Wildcards



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Managing Client Connections : Transport Connectors

- Configured in broker for client connections

- TCP – most used; socket connections using binary Openwire protocol
- NIO – like TCP, excepts uses Java NIO to reduce number of threads managing all connections
- SSL – secure TCP connection
- STOMP – text based protocol; facilitates multiple language integration
- VM – enables efficient in-process connections for embedded broker

- Examples
  - <transportConnector uri="tcp://0.0.0.0:61616"/>
  - <transportConnector uri="nio://0.0.0.0:61616"/>
  - <transportConnector uri="stomp://0.0.0.0:61617"/>
  - <transportConnector uri="stomp+nio://0.0.0.0:61617"/>

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Managing Client Connections : Wrapper Transports

- Augment / wrap client side connections
- Failover – automatic reconnection from connection failures
- Fanout – simultaneously replicate commands and message to multiple brokers
- Example – client connection URI
  - tcp://master:61616
  - failover:(tcp://master:61616,tcp://slave:61616)
  - failover:(tcp://virtualIp:61616)
  - fanout:(static:(tcp://host1:61616,tcp://host2:61616))

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Managing Client Connections : Configuring Transports

- tcp://hostname:port?key=value

- Examples
  - tcp://myhost:61616? trace=false&soTimeout=60000
  - failover:(tcp://master:61616?soTimeout=60000,tcp://slave:61616)?randomize=false

- Lot more details at
  - http://fusesource.com/documentation/fuse-message-broker-documentation/
  - http://activemq.apache.org/configuring-transports.html

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Managing Persistence : Persistence Adapters

- File system based
  - kahaDB – recommended; improved scalability and quick recovery
  - amqPersistenceAdapter – legacy; fast, but slow recovery
- RDBMS based
  - jdbcPersistenceAdapter – quick and easy to setup
  - journaledJDBC – faster than pure JDBC; file journaling with long term JDBC storage
- Memory based
  - memoryPersistenceAdapter – testing only; same as
    - <broker persistent="false">

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High Availability

- Two complementary approaches:
  - Master/Slave – access to persistent messages after broker failure
  - Network of Brokers – Scale out message processes - next slides…

- Master/Slave Context
  - A given message is in one and only one broker (persistence store)
  - If a broker instance fails, all persistent messages are recoverable upon broker restart
  - Master/Slave allows a 2nd broker instance (slave) to be ready to process persistent messages upon master (1st broker) failure
  - Clients should use Failover transport for automatic connect to slave
    - failover:(tcp://master:61616,tcp://slave:61616)?randomize=false

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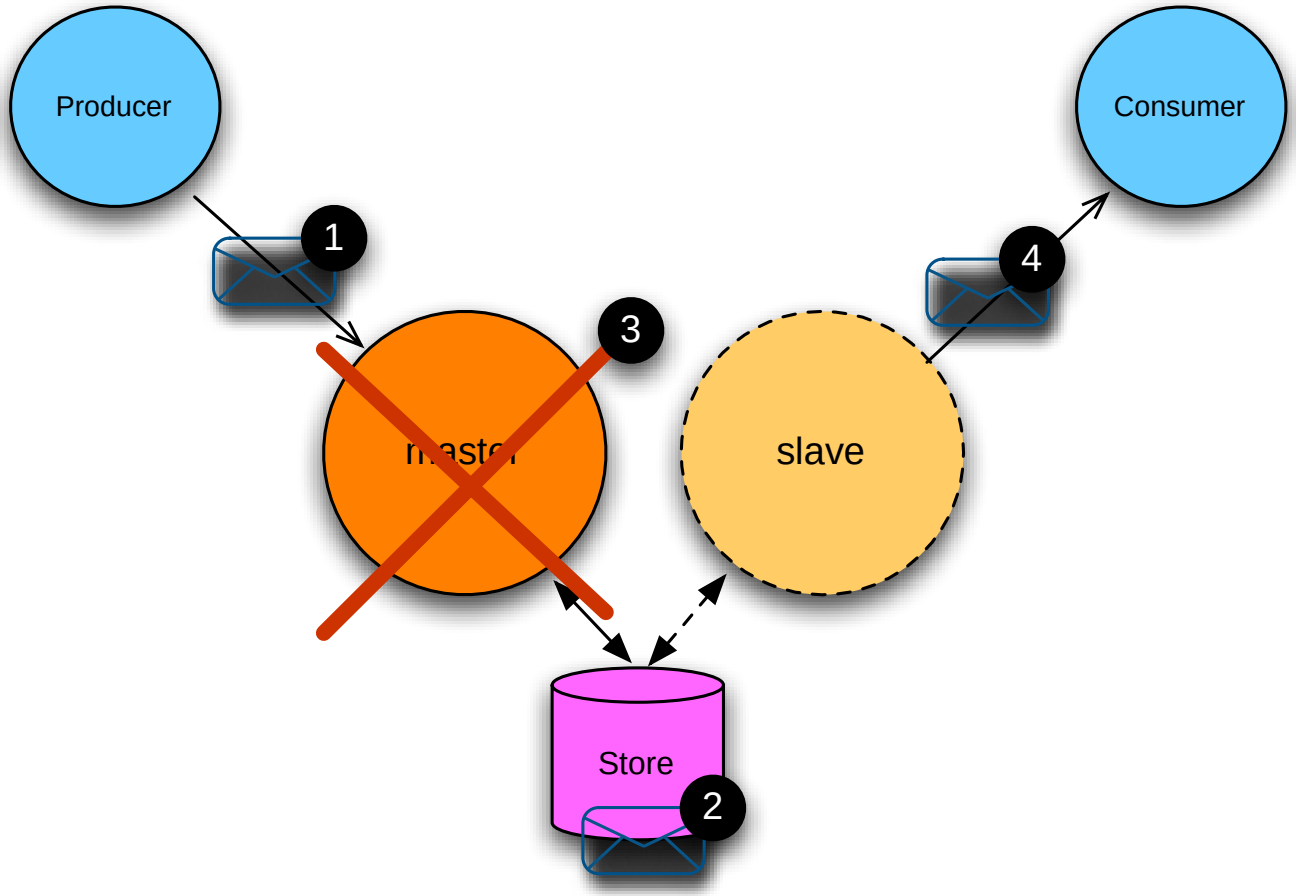
High Availability : Master/Slave



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Network of Brokers : Geographically Dispersed

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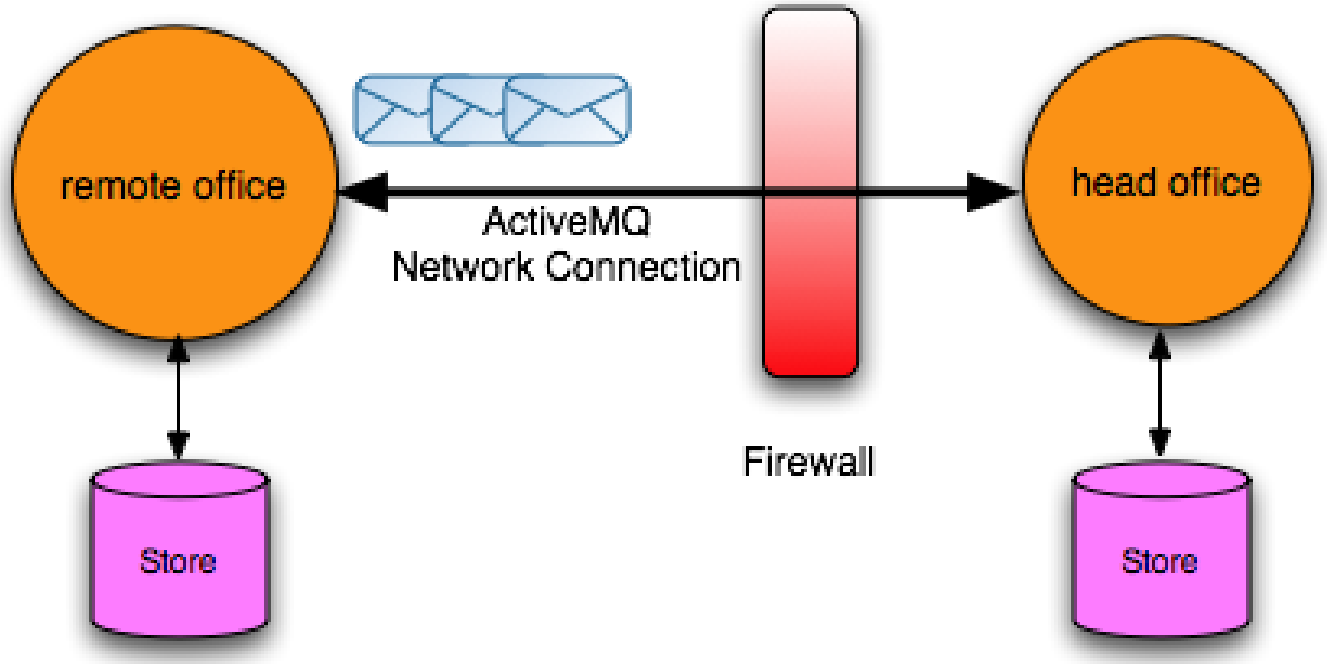
Network of Brokers : Geographically Dispersed



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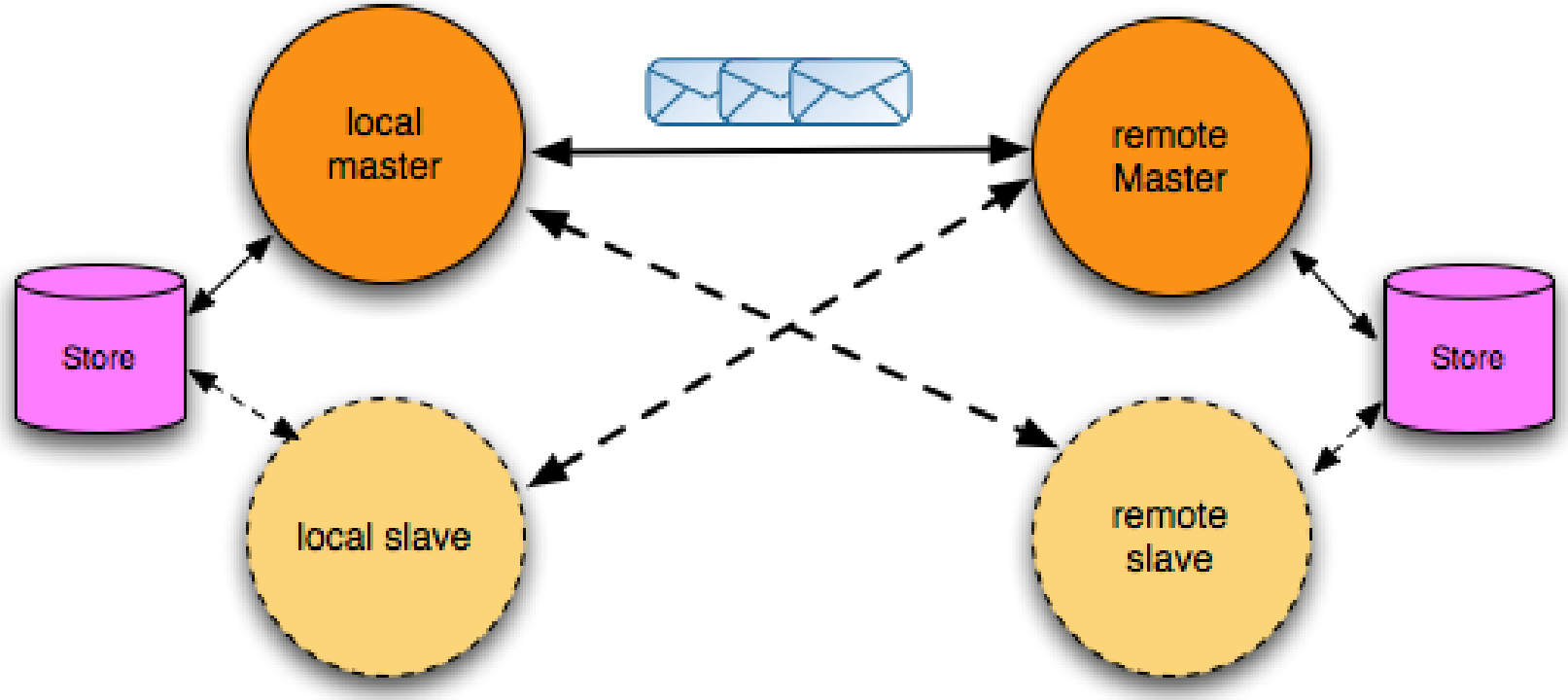
Network of Brokers : Network with Master/Slave



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Code Time

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Great Stuff at FuseSource.com

If you missed ***CamelOne*** *some video's and presentations can be found on the FuseSource site.*

Virtual Training:

Developer Training for ActiveMQ USA & EMEA dates in Sept and Nov

Developer Training for ServiceMix 4.x with Camel USA & EMEA Dates in Sep & Nov

FuseSource Community Day's

October: UK & France

November: Chicago & San Francisco

Come and meet our some of our engineers like James Strachan and Rob Davies as well as our customers talk about the benefits of using Open Source with your integration and messaging projects.

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